Bakehouse Retail Analytics Dashboard

Executive Summary: Bakehouse Retail Analytics

Project Overview: Implemented an end-to-end analytics solution for Bakehouse Retail, establishing a medallion architecture data pipeline and interactive dashboard to analyze 200 transactions across 6 store locations over a 28-day period (January 6 - February 2, 2025).

Key Business Insights

Revenue Performance: Generated $18,626.64 in total revenue across 200 transactions, with an average transaction value of $93.13. The business moved 625 total items during the analysis period.

Product Strategy: Electronics dominates revenue contribution at 42% ($7,852), followed by Apparel (24%, $4,410). Running Shoes emerged as the clear winner with $3,600 in sales representing 19% of total revenue. The top 4 products (Running Shoes, Mechanical Keyboard, Backpack, Headphones) account for over half of all revenue.

Store Operations: Orlando leads store performance with $4,291 in sales, while Nashville lags at $1,841—a $2,450 gap indicating potential operational disparities or market differences requiring investigation.

Customer Behavior: Cash remains the preferred payment method (27% of transactions, $5,034), suggesting a customer base comfortable with traditional payment. Monday drives the strongest sales ($3,938, 21% of weekly revenue), while Tuesday underperforms significantly ($2,093, 11%).

Growth Trajectory: Weekly revenue showed positive momentum from Week 2 ($4,251) to Week 4 ($4,966)—a 16.8% increase—though Week 5 experienced a slight pullback to $4,875.

Total Revenue

$18.63K

Total Transactions (Units)

200

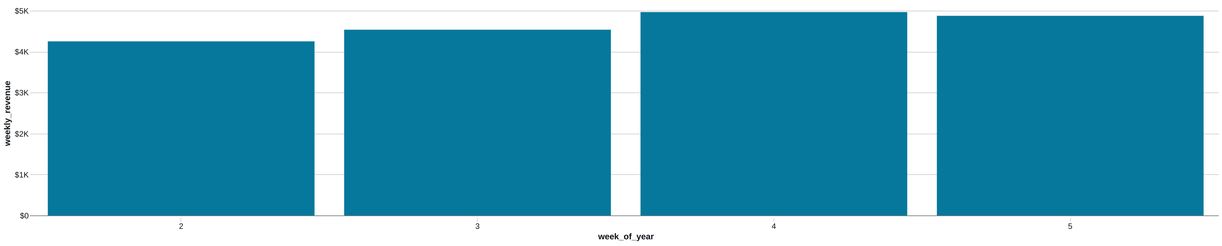
Avg Transaction Value

$93.13

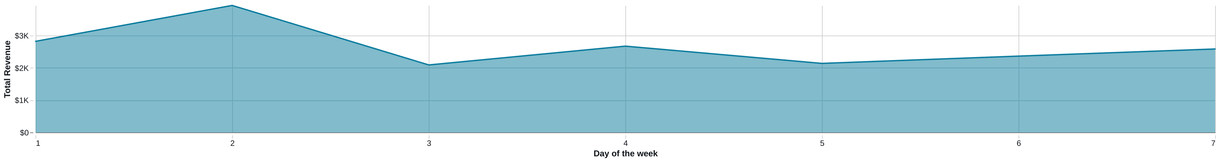
Total Items Sold (Units)

625

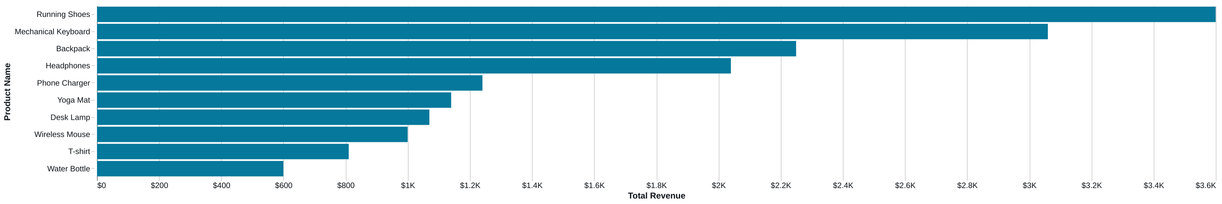
Weekly Revenue Trend



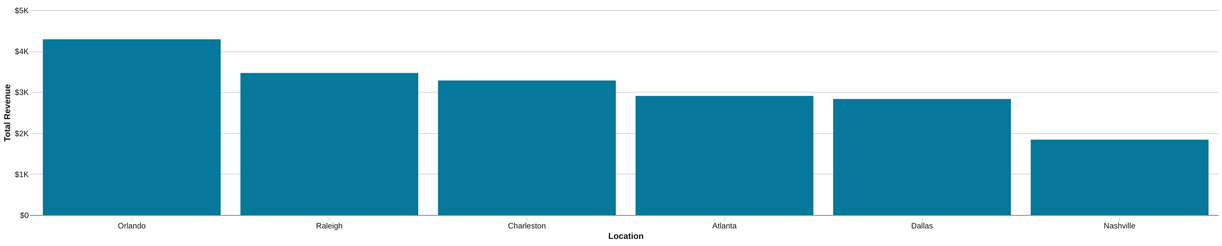
Sales by Day of Week



Top 10 Products by Revenue



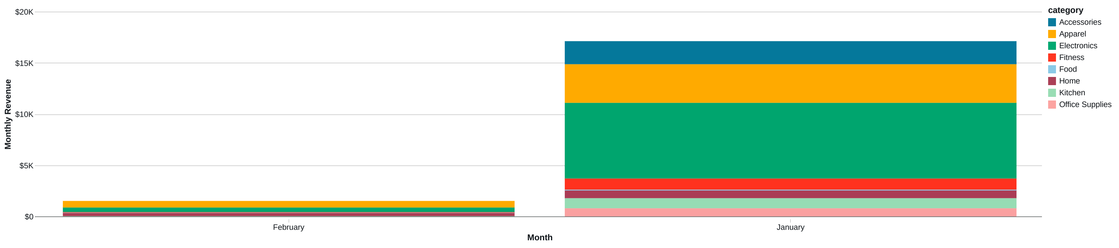
Store Location Performance



Category Performance

Payment Method Distribution

Monthly Sales by Category



Strategic Recommendations

- ***Store Optimization:*** Investigate Nashville's underperformance relative to Orlando. Consider operational audits, staffing levels, or localized marketing campaigns.
- ***Inventory Focus:*** Prioritize Electronics and top-performing products (Running Shoes, Mechanical Keyboard) while evaluating lower-tier category performance.
- ***Weekday Strategy:*** Capitalize on Monday's strong performance with targeted promotions. Address Tuesday's weakness through special offers or operational adjustments.
- ***Payment Modernization:*** While cash dominates, digital payment adoption (Google Pay, Apple Pay) remains low—opportunities exist for contactless payment incentives.

Technical Foundation: Infrastructure & Data Quality Summary

Medallion Architecture Implementation

Bronze Layer (Raw Ingestion)

- ***Location:*** `/Volumes/bakehouse_retail/bakehouse_retail_bronze/raw/`
- ***Source Files:*** 4 file types ingested
- `customersjson.json` (5 customers)
- `ordersjson.json` (5 orders)
- `productsjson.json` (4 products)
- `transactions_*.csv` (200 transactions)
- ***Catalog/Schema:*** `bakehouse_retail.bakehouse_retail_bronze`
- ***Purpose:*** Immutable landing zone for raw source data

Silver Layer (Data Cleaning)

- ***Catalog/Schema:*** `bakehouse_retail.bakehouse_retail_silver`
- ***Tables:*** 4 cleaned tables (`customers_clean` , `orders_clean` , `products_clean` , `transactions_clean`)
- ***Transformations Applied:***